

AI-Based Resume Screening and Job Role Classification

1. Problem Statement

Recruiters and placement teams receive a large number of resumes for different job positions. Manually reviewing every resume and identifying the most suitable job role can be time-consuming and inconsistent.

The proposed AI solution analyzes the textual content of a resume and classifies it into a suitable job category based on the candidate's skills, education, experience, and projects.

The initial system will focus on a small set of job categories:

- Software Developer
- Data Analyst
- Data Scientist
- Web Developer
- Java Developer

The goal is to demonstrate how AI-based text classification can assist with the initial screening of resumes.

2. Target Users

The primary users of the proposed system are:

- HR professionals
- Technical recruiters
- College placement officers
- Recruitment teams
- Job seekers who want to understand which role their resume best matches

The system is intended to assist with initial screening and should not be used as the sole basis for making hiring decisions.

3. Data Source

A small labeled resume dataset will be used for the initial experiment.

Each record contains resume-related textual information such as:

- Technical skills
- Educational background
- Work experience
- Projects
- Certifications
- Professional summary

Each resume is assigned a predefined job-role label.

Example:

Resume Information	Job Role
Java, Spring Boot, MySQL, REST API	Java Developer
Python, Pandas, SQL, Power BI, Excel	Data Analyst
Python, Scikit-learn, TensorFlow, ML	Data Scientist
HTML, CSS, JavaScript, React	Web Developer
Java, DSA, OOP, Spring Boot	Software Developer

For privacy and ethical considerations, personally identifiable information such as names, phone numbers, email addresses, and physical addresses should be removed from the dataset.

4. AI Problem Type

Problem Type: Supervised Machine Learning – Text Classification

Input

A candidate's resume text.

Output

A predicted job category.

Example:

Input:

Python, SQL, Pandas, NumPy, Power BI,
Excel, data visualization and dashboard development.

Output:

Data Analyst

5. Proposed AI Approach

The system can follow these steps:

Resume



Text Extraction



Text Preprocessing



Feature Extraction



Classification Model



Predicted Job Role

Step 1 – Resume Text Extraction

Extract the text from the uploaded resume.

Possible file formats:

- PDF
- DOCX
- TXT

Step 2 – Text Preprocessing

The extracted text will be cleaned by:

- Converting text to lowercase
- Removing unnecessary symbols
- Removing irrelevant whitespace
- Handling common stop words where appropriate
- Normalizing the text

Step 3 – Feature Extraction

Text can be converted into numerical features using techniques such as:

- TF-IDF
- Word embeddings
- Sentence embeddings

For the initial prototype, **TF-IDF** can be used because it is simple, interpretable, and suitable for a small dataset.

Step 4 – Classification

A classification algorithm can be trained to predict the job category.

Possible models include:

- Logistic Regression
- Naive Bayes
- Support Vector Machine
- Random Forest

For the initial experiment, Logistic Regression or Support Vector Machine can be evaluated.

Step 5 – Prediction

The trained model receives an unseen resume and returns the predicted job category.

6. Constraints

The system has several practical constraints:

1. The initial dataset will be relatively small.
2. Resume formats and writing styles can vary considerably.
3. Candidates may use different names for similar skills.
4. Some resumes may be suitable for multiple job roles.

5. Missing information can affect classification.
6. Dataset imbalance may affect model performance.
7. Personally identifiable information must not be used unnecessarily.
8. The system should assist recruiters rather than automatically make final hiring decisions.

7. Success Criteria

The AI system will be considered successful if it achieves the following target metrics on unseen test data:

Metric	Target
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Accuracy	$\geq 80\%$
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Precision	$\geq 80\%$
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Recall	$\geq 80\%$
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F1-Score	$\geq 80\%$
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Additional practical criteria:

- The system should produce a prediction within a few seconds.
- Each job category should have reasonable classification performance.
- The model should be tested on resumes that were not used during training.
- Incorrect predictions should be analyzed to identify common failure cases.

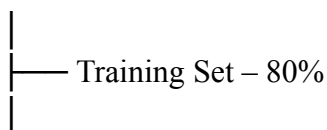
The 80% values are defined as **project targets**, not guaranteed results.

8. Evaluation Approach

The dataset will be divided into training and testing subsets.

Example:

Dataset



└── Testing Set – 20%

The training set will be used to build the classification model, while the testing set will be used only for final evaluation.

The following metrics will be calculated:

Accuracy

Measures the percentage of correctly classified resumes.

Precision

Measures how many resumes predicted as a particular role actually belong to that role.

Recall

Measures how many resumes belonging to a particular role were correctly identified.

F1-Score

Provides a balance between precision and recall.

A confusion matrix will also be used to identify which job categories are frequently confused with each other.

9. Example Evaluation

Suppose the model is tested on 100 unseen resumes.

An example target evaluation could look like:

Metric	Result
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Accuracy	84%
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Precision	83%
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Recall	82%
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F1-Score	82.5%
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These numbers are examples of how the final results can be reported. Actual values should only be added after running the model on the dataset.

10. Expected Benefits

The proposed AI system could help:

- Reduce the time required for initial resume screening.
- Categorize resumes consistently.
- Assist recruiters in organizing candidates.
- Help placement teams handle large numbers of student resumes.
- Provide job seekers with an indication of which role their current skills align with.

11. Limitations

The system will not determine whether a candidate should be hired.

A resume is only one source of information and may not accurately represent a person's complete skills or potential.

Other limitations include:

- Small datasets may not generalize well.
- Similar job roles can have overlapping skills.
- Resume quality can affect predictions.
- The model may inherit biases present in its training data.
- Confidence scores should not be interpreted as hiring probabilities.

Human review should remain part of the recruitment process.

12. Future Scope

The project can later be extended with:

- Resume-to-job-description matching
- Skill-gap analysis
- Job recommendation

- Semantic embeddings
- Large Language Model-based analysis
- Explainable AI for predictions
- Multiple resume format support
- Recruiter dashboard
- Candidate ranking based on explicit job requirements

13. Conclusion

This AI problem demonstrates how supervised text classification can be applied to a practical recruitment use case.

The proposed system takes resume information as input and predicts a suitable job category. Its performance will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

The initial objective is not to automate hiring decisions, but to demonstrate how AI can support the early stage of resume organization and screening while keeping human judgment in the process.